

The Promise and Challenge of Large Language Models for Knowledge Engineering: Insights from a Hackathon

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ABSTRACT

Knowledge engineering (KE) is the process of building, maintaining and using knowledge-based systems. This recently takes the form of knowledge graphs (KGs). The advent of new technologies like Large Language Models (LLMs) has the potential to improve automation in KE work due to the richness of their training data and their performance at solving natural language processing tasks. We conducted a multiple-methods study exploring user opinions and needs regarding the use of LLMs in KE. We used ethnographic techniques to observe KE workers using LLMs to solve KE tasks during a hackathon, followed by interviews with some of the participants. This interim study found that despite LLMs' promising capabilities for efficient knowledge acquisition and requirements elicitation, their effective deployment requires an extended set of capabilities and training, particularly in prompting and understanding data. LLMs can be useful for simple quality assessment tasks, but in complex scenarios, the output is hard to control and evaluation may require novel approaches. With this study, we aim to evidence the interaction of KE stakeholders with LLMs, identify areas of potential, and understand the barriers to their effective use. We find copilot approaches may be valuable in developing processes where the human or a team of humans is assisted by generative AI.

CCS CONCEPTS

• **Computing methodologies** → **Ontology engineering**; • **Human-centered computing** → *Natural language interfaces*.

KEYWORDS

Knowledge Graph, Knowledge Engineering, Large Language Models, Interviews

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1 INTRODUCTION

Knowledge engineering (KE) is the process of capturing, structuring, representing and maintaining knowledge in a machine-readable way [57]. The outcome of this process often takes the form of a knowledge graph (KG) [24]. An example is the Google KG, familiar to many as the factual information that appears in the automatically-generated side panel on search results pages [56]. Historically KE is a manual, curated process, hence scaling is a major challenge [47, 51]. Automation has been used to support this [33, 35]. Recent developments in large language models (LLMs) have enhanced KE automation throughout large parts of the life cycle [4, 51].

There are a number of existing challenges in KE [51]. Two of these are:

- Knowledge acquisition and requirement elicitation focuses on modelling and constructing knowledge graphs by extracting and forming entities, relations and attributes from various sources and stakeholders [51]. However, this process can be costly, inaccurate and time consuming, especially where expert domain knowledge is required [3, 51]. Inadequate knowledge acquisition results in content gaps in KGs that impact downstream applications and can be harmful [2].
- Quality assessment is necessary to ensure the 'fitness for purpose' or 'fitness for use' of a KG [10, 26]. Quality assessment in KE is essential to accurately modelling a domain in order to be a trusted, curated, machine-readable representation of knowledge. A standard practice is to ensure the KG accuracy, coverage, coherency and succinctness. The process demands domain expertise, manual work and time. While semi-automatic evaluation solutions exist, they are not yet sufficient [51].

One more important aspect of KE is multimodality. Multimodal KGs attempt to improve "real-world" representation by including formats such as natural language text, images, videos, and audio [43, 58, 73]. Selection, annotation and labelling of multimodal content is intensive, mixing modalities adds layers of challenge and inefficiency [51]. Multimodal LLMs could be useful for KE tasks

when they are mature enough, which has seen substantial advances recently.

Machine learning and natural language processing techniques have been used to enhance KE, with LLMs creating substantial performance improvement in tasks such as entity recognition, entity relation and error correction [15, 33]. LLMs have also introduced new capabilities for supporting complicated tasks such as ontology design or multimodality. However, despite the affordances of LLMs there are still challenges to their use: their training data often has a cut-off date and their output sometimes lacks credibility. There are many initiatives to evaluate LLMs broadly, such as HELM [6], however, evaluating LLMs for specific tasks requires substantial effort in setting up benchmarks and potentially developing focused evaluation techniques [4] as well as mapping them to existing notions of quality.

To understand how those involved in KE view the perils and promises of using LLMs, we devised a multi-method qualitative study [41]. This was centered around a four-day hackathon exploring the use of LLMs to address key topics in KE. Our initial study answers the following research questions:

- How do people working in knowledge engineering perceive the use of LLMs to support KE tasks?
- What do users believe to be the limitations of LLMs in these contexts?

2 BACKGROUND WORK

In this section, we define KE, present more detail on key challenges in the construction of KGs and document the use of LLMs in KE.

2.1 Knowledge Engineering

Early KE focused on developing Knowledge Base Systems [57]. These have been superseded by graphs. These KGs represent real-world entities (nodes) and their relations (edges) [24]. The basic unit of the KG is the triple – subject, predicate, object. Ontologies can provide a structure and meaning behind the KG. This formalisation of the representations through semantics enables them to be processed computationally. The graph structure offers efficient data management, availability of data in different modalities, use of machine learning and crowdsourcing techniques, and lower cost [47]. KG construction activities depend on the actors involved, the represented domain, the construction tools, and the available data source [24]. Key actors generally include domain experts, who specialise in the intricacies of the topic under consideration. Knowledge engineers elicit these insights to create a conceptual domain model, and ontology engineers represent this knowledge in a suitable machine language [55]. KGs can be built and enriched using human collaboration (e.g. crowdsourcing platforms) [22, 62], and automatic techniques (e.g. natural language processing) [37, 38] to extract knowledge from text, markup/HTML, and structured sources (e.g. tables, JSON, XML). Competency questions, i.e. natural language questions representing the domain information required [32, 55], can be used to assess the completeness of the represented domain. It is not currently possible to fully automatically construct a KG [23].

2.2 Key Issues in Knowledge Engineering

Knowledge acquisition and requirements elicitation. Knowledge acquisition can be manual or automated, imported from multiple sources, structured and unstructured. The world’s largest KG is Wikidata with more than 108M entities, which is built using manual and automated techniques by a community of volunteers [67]. The process requires time and human effort, with the community being responsible for maintaining and growing the graph, preventing vandalism, and ensuring high quality. Manual acquisition is time-consuming and costly. Automated acquisition depends on advanced technology and large-scale datasets. Automatically constructed KGs like DBpedia [14] with 6M or AceKG [65] with 114M entities differ from the manually curated Wikidata in size and domain complexity. Imperfect knowledge acquisition may result in low accuracy, which can result in low-quality KGs and thereafter negatively impact downstream tasks [51].

Quality assessment. Quality assessment is the evaluation of the KG on a number of dimensions [24]. KGs must be evaluated on their syntactic and semantic accuracy, and their timeliness [18, 19, 28, 37]. Coverage is the KG value of completeness and representativeness (assessing the level of bias of the existing knowledge in the KG, whether this is inherent in the data, the schema or the reasoning) [13, 27, 29, 37]. Coherency refers to consistency - whether the KG is free of contradiction and consistent with the domain it represents - and validity [7, 59]. Succinctness refers to the KG’s conciseness and understandability (by humans) [25, 29].

The above categories can be assessed based on their type using validation tools [19] such as human evaluation; checking the relations and measuring precision and recall against gold standards or other sources; or comparing known statistical distributions with those of the KG [13, 18, 37]. While not complicated, these assessments can involve a considerable amount of effort. A standard practice to ensure quality and to correct errors is to use rules and constraints that the KG needs to follow [24]. Domain experts are the most suitable specialists for creating these rules, however this is an expensive, manual process.

Other Challenges. These are not by any means the only current challenges in the field. Multimodality in KGs provides structured representation of data of multiple types. This facilitates machine perception of scenarios and thus improves performance [64]. Images, text, video, audio and haptics of the same entity can complement and enhance each other, which leads to richer and more complete KGs [12]. These can be used to represent knowledge about highly complex situations, such as smart cities or international conflicts [34, 60]. Some knowledge can only be accurately represented in its native modality, for example, musical chords [16]. Additionally, while humans are good at providing textual input, some tasks require output in other modes, such as haptics [44]. The extensive availability of images of the web means that visuals are the most often used media in KGs [12]. However, finding and integrating other multimodal information can be challenging. There are limited examples of multimodal datasets, making it difficult to find data and ensure that the different modalities refer to the same entity [12]. These issues are compounded in specific verticals. Knowledge extraction for domains such as heritage remains complicated and time-consuming [9, 16, 61].

Representativeness touches on bias, but this is only part of a far bigger issue of responsibility and safety in KGs. Non-English training datasets are not abundant, translation systems are not infallible and entity extraction models must be retrained in each language, making multilingual KG construction far from easy [51]. A problem that will also be familiar to many, is that of entity disambiguation (the 007 James Bond/007 Nomi problem) and entity matching (the James Bond, Universal Exports/James Bond, 007 problem) [52, 72]. During our research we collected further data pertaining to these aspects and intend to explore them as our research develops. The three issues we are focusing on initially are problems which depend greatly on automating at scale to address. The potential for this has been increased in recent years by the advent of LLMs and in the next section, we review how these have contributed to and problematised KE.

2.3 Large Language Models and Knowledge Engineering

KG construction has changed dramatically over the years using semi-automated processes relying on deep-learning models and vast collections of heterogeneous data sources to scale [69]. The deep-learning models have been used to support KE for knowledge extraction, KG refinement, and KG enrichment. Knowledge extraction i.e., the knowledge derived from structured and unstructured sources [47] requires manual and automated approaches (supervised and unsupervised) from the respective domain. LLMs are used to: identify named entities in a text; linking people, organisations, and locations [66] to associated entities in a KG with mentions of entities in text [30]; and to identify the relation between entities in a text [8]. KG refinement refers to techniques for KG completion and correction [33, 35]. For completion, the aim is to add missing knowledge to the graph. Automatic methods using LLMs fill in missing edges between KG entities and add new entities [35]. In contrast, for correction, the aim is to correct existing entities and relations. KG corrections are possible using LLMs for evaluating whether relations are true or using reasoning techniques examining the ontology axioms to identify inconsistencies [51]. KG enrichment can incorporate ontology refinement and alignment to improve a given KG. In refinement, the aim is to improve text inconsistencies and add meta information like descriptions to entities [35]. In addition, for alignment, LLMs have been used to identify and match entities between KGs, for example, using lexical matching [70]. Figure 1 shows the KG construction outline, related to key LLM areas of use as explored in the hackathon topics.

Even using the most recent advances in natural language processing [42], the challenge of how to capture and encode domain knowledge into a computational representation remains [69]. KE tasks like ontology alignment and design using pre-trained models have been tested [11, 46] and shown to improve performance, yet still lack sufficient credibility to be used without human supervision. A lack of credibility is amplified by the lack of provenance associated with LLM output and their well-documented tendency to hallucination [5]. Therefore, when LLMs are introduced, trust is impacted. Trust is a vital aspect of KGs. This means we need new assurance methods that take these issues into account.

Despite the challenges and barriers to co-opting LLMs for KG construction, this improved technology also offers capabilities, such as enhancing multimodality. The advent of AI systems that can create realistic images [48] or even videos [49] is still evolving but they offer great potential for advancing multimodal KGs.

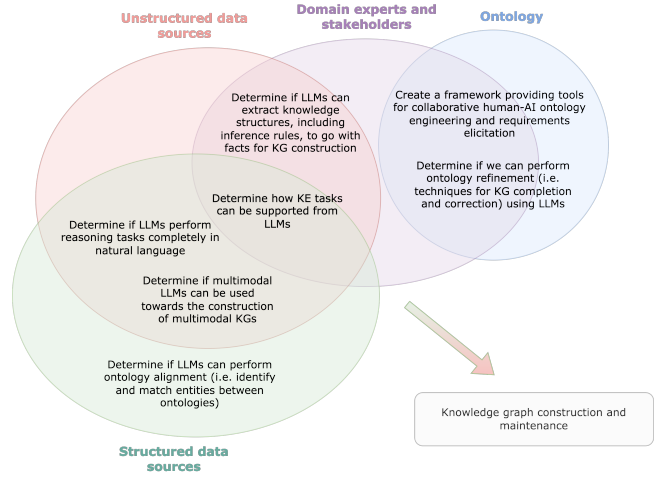


Figure 1: KG construction outline including the hackathon topics.

3 METHODOLOGY

During August 2023 we attended a four day research hackathon aimed at bringing together researchers and practitioners interested in using the latest advances in LLMs and generative AI to create and maintain curated, credible, and trustworthy KGs [31]. Attendees were selected from applicants based on their background and experience in KE. They worked in European research labs with a strong profile in either publishing in KE venues, or in offering popular KE industry products. In total the 39 participants were from 15 different institutes and various environments from academia and industry, with 31 PhD students, 2 postdocs, 1 lecturer, 2 professors, and 3 industry members. Participants split into 7 groups of 5 to 7 members, based on their prior experience in KE. Each group investigated one of the topics presented in Table 1.

3.1 Research Approach

The overall research programme was devised to assess how effectively LLMs can contribute to the KE lifecycle. Rather than a large-scale automated approach, which can be seen in e.g. [71], we wished to understand participants' opinions of the experience of interacting with LLMs in this evolving field in a range of situations. Thus a qualitative approach was selected. This paper represents a subset of the full research, focusing on the opinions of participants regarding the use of LLMs to address issues of knowledge acquisition, multimodality and quality assessment. Particular findings and insight regarding the capabilities of LLMs in each challenge listed above will be presented in a further paper.

Table 1: Description of hackathon topics with associated problem areas.

Hackathon topic	Problem area addressed	KE Tools/Methodologies	LLM and Tools/Methodologies
Determine if LLMs can extract knowledge structures, including inference rules, to go with facts for KG construction	knowledge acquisition, Requirements elicitation, Multimodality	Triples investigation	ChatGPT, Few-shot prompt
Create a framework providing tools for collaborative human-AI ontology engineering and requirements elicitation	Knowledge acquisition, Requirements elicitation, Quality assessment	Competency Questions, eXtreme Design methodology	ChatGPT, Multiple prompting techniques
Determine how KE tasks can be supported from LLMs	Knowledge acquisition, Requirements elicitation, Quality assessment	NeOn methodology, Competency Questions, HerMiT Reasoner, OOPS	PaLM, Llama, ChatGPT, Few-shot prompt
Determine if LLMs perform reasoning tasks completely in natural language	Reasoning	Triples investigation	ChatGPT, Multiple prompting techniques
Determine if LLMs can perform ontology alignment (i.e. identify and match entities between ontologies)	Quality assessment, Knowledge reuse	OAEI	ChatGPT, Zero-shot prompt
Determine if multimodal LLMs can be used towards the construction of multimodal KGs	Multimodality	Triples investigation	mPLUG-Owl, InstructBLIP, Text only prompt (no text + image)
Determine if we can perform ontology refinement (i.e. techniques for KG completion and correction) using LLMs	Knowledge acquisition, Quality assessment	OntoClean	ChatGPT, Llama, Claud, Few-shot prompt

3.2 Data Collection

During the hackathon, an ethnographic observation study of the participants working on the challenges in groups was undertaken by two of the authors, using the “observer as participant” technique [17]. After the hackathon, we contacted 14 attendees who had volunteered to participate in semi-structured interviews. Interviewees had backgrounds in KG construction, KG explainability, ontologies and machine learning, as well as biology and bioinformatics, data management, FAIR data [68], and digital humanities. We conducted these interviews virtually between 17/8/2023 and 6/9/2023 using Microsoft Teams [40]. The interview guide can be found in Appendix A. We used Otter.ai [50] to transcribe the interview recordings.

We further gathered research output from the hackathon in the form of reports and documentation for documentary analysis (14 documents). The observations and reports informed the design of the interview, and will contribute to answering further RQs in the extended research. This and the interview transcripts were then uploaded into NVIVO [36] for analysis.

3.3 Data Analysis

We conducted an initial thematic analysis across data from the interviews and ethnography. We used an inductive method to search for key themes from the interview topic guide (see Appendix A). We then used a deductive approach to identify grounded themes that emerged in the data. While we had previously identified issues with quality assessment and therefore specified these in the interviews, the issues with multimodality and knowledge acquisition were not previously identified and arose during the deductive analysis. The analysis was performed by two researchers, who first read the interview topic guide and together identified the key themes. They then read the data corpus for the grounded themes. Following this, the two researchers, advised by a senior researcher, discussed and confirmed the codes.

3.4 Ethics

The study was approved by the authors’ institution’s Ethical Advisory Committee via the Full Application Form. Informed electronic (using Microsoft Forms [39]) and verbal consent was given by the participants. Participants who did not consent were not observed. No personal information was used in the analysis.

4 RESULTS

Our larger study will eventually address what the participants were able to achieve when developing outputs in the hackathon using

LLMs for KE. These interim results focus mainly on participants’ experience of co-creating with LLMs and how that has informed their views on future use of LLMs in KE.

4.1 Knowledge Acquisition and Requirements Elicitation

Interviewees (1), (14), (12) emphasised the importance of datasets as a starting point. A key concern was around the trustworthiness of data provided by LLMs. One such view was, “Yes, it can help you to quickly get some data just for testing purposes, for example. But for production, I don’t think that I will use it...since I never know which part of it is actually correct.” (6). Groups used prompting of LLMs as a means to create a dataset (Observer 1). A lack of a dataset presented a major barrier to continuing prompting and evaluation (14). “There are many codes online...to interact with language models, that wasn’t any problem at all. And the evaluation given that you know, you possess a ground truth...was not too difficult. So, the most challenging case work for us was finding a dataset.” (8). Finding a dataset was challenging for specific group tasks mainly because of the time pressure (14) and fast planning (1) needed to find and set domain knowledge for the task. However, an interviewee noted that they felt confident LLMs could support knowledge acquisition. “The dataset is not what concerns me because if anything, I’m pretty impressed with what LLMs potentially could do in terms of extracting, concepts, relationships, maybe even constraints or parameters or universal restrictions or existential restrictions, from unstructured data.” (13). However, it is worth noting that the same interviewee expressed that integrating KGs and LLMs may not be required as LLMs could potentially entirely replace KGs. “What’s the point of using LLMs to create KGs or KE because one can ask the question directly of [an] LLM, can’t they,” (13). One of the key aspects the users found when attempting to acquire knowledge was developing suitable prompts. One interviewee believed prompting is very training and experience dependent. “I think prompting is going to be generally more difficult for Semantic Web people, because we are not necessarily natural language processing people” (1). Furthermore, many participants noted the time-consuming phase of iteratively testing prompts to promote desirable outcomes (3), (Observer 1). While a prompt may produce the correct results initially, “the most tricky part of it was to make the language model consistent in its responses” (3). Another noted

that prompting engineering is “less scientific” and models “with billions of parameters are not controllable..., resulting in an output that you cannot reproduce one hour later, two days later” (9). This was particularly problematic when the process was automated with thousands of prompting iterations. Despite the lack of consistency in prompting outcomes, some suggested the usefulness of possible “templates for prompting” to support those with less experience in natural language processing (Observer 1), (Observer 2), (3), (13). Additionally, there was concern that syntactical errors created by the LLMs may break scripts and prevent automation when using more unstructured data (3).

4.2 Quality Assessment

Competency questions are used in quality assessment to identify whether a KG is complete for a particular domain. Two interviewees (3), (6) felt that LLMs offered promise as assistants making this evaluation smoother and faster because, “we know that as humans, we are lazy to write down some stuff like the competency question, so, if [you have several] competency question(s) you decide [faster], actually, which one you want to use” (6). They suggested that LLM use should focus on human satisfaction, “so the language model should probably optimise not really the performances, but the satisfaction of the user” (3). One interviewee (5) suggested that we need to be more “open-minded” in order to advance the field of quality assessment using LLMs. If metrics are, “novel, it takes longer to review. If it’s a straightforward paper with a slight modification in the method and a new row, it’s easier to evaluate the merit of the work. That’s not necessarily the way to go. So there’s a lot of work to be done in evaluation” (5). Furthermore, another factor in output evaluation is the similarity of the results (4). Output is considered as false even if it is close to or similar to the truth. The probabilistic element of LLMs was therefore considered a barrier.

While quality assessment focuses on the end output of a KG, participants also were aware of challenges with evaluating the quality of LLM-assisted individual components of knowledge graphs. Some focused on the ontology design (2), (5), suggesting the development of a set of ontologies to be used as gold standards (Observer 1). Others suggested the creation of toolkits (1) for LLMs to review ontology errors similar to existing ontology toolkits, like OntoClean [21] and OOPS [53]. However, they acknowledged that this cannot be applied to large-scale KG like Wikidata, and manual work will still be required. Even if we did have the capacity to automatically evaluate the “generative text”, LLM outputs are not all the same type (4), for example, a KG includes dates, values, images, URLs etc. Interviewees felt it is not possible to have a united way to evaluate the different types.

4.3 Other Tasks

Overall participants felt that LLMs did improve the potential for multimodal KGs and were excited about this. “We were really optimistic [that] multimodal models could exist...that we can insert into the models more than just image on text, but also sounds etc” (2). But it also raised awareness of the increased human skills that would be needed. For example, “we evaluate three different kinds of extractions, one just based on tags we want, [one] just based on images [and]...one was based on images and texts” (4). Observer

2 noted that a group found the setup of the prompting for working with images was particularly hard. LLMs did not seem to offer simplification of the challenges of multimodality. One participant expressed that, “issues were exacerbated during the hackathon because we were looking for multimodal language models. And that was a big thing” (5).

4.4 Summary

- Knowledge acquisition and requirements elicitation: LLMs show promise but are not trusted, and their use for knowledge acquisition is dependent on effective prompting, which is still subject to inconsistency.
- Quality assessment: LLMs can be useful for creating competency questions, and other simple tasks, but in complex scenarios, the output cannot be controlled. Therefore, humans need to stay in the loop, and also consider new forms of evaluation.
- Multimodal KGs: a barrier to use of LLMs is the need for both LLM and multimodal skills in order to use these effectively.

5 DISCUSSION

Trustworthiness and reliability are the keystone of KGs, and as such they have been used for grounding LLM results through retrieval-augmented generation. However, when reversing the relationship, a fundamental conflict occurs in the use of LLMs in KG work because of the ambiguity inherent in the output. This ambiguity lies mostly in the inherent irreproducibility of results garnered using generative AI. This means that contributions made by LLMs, such as the creation of relations or the similarity between terms, are based on statistical probability rather than consistent fact. This lack of consistency is also problematic for scientific method. This is consistent with previous studies arguing that knowledge extraction remains challenging despite the growing field of natural language processing [4, 47]. However, one notable use of the LLMs for KE is to potentially create a structured dataset from a variety of unstructured data. Other research [63] has found that given a prompt that includes the term “dataset”, in the absence of a dataset, LLMs may attempt to “curate” a dataset like approximation from unstructured data. Therefore, this application seems highly plausible for KE and, particularly, knowledge extraction in a structured format which is similar to existing practices. Following the usual actions of cleaning and transforming, the dataset can be used for KG construction. Yet, the accuracy of LLM results remains an issue.

Attempting to evaluate KGs with LLM support will simply exacerbate existing evaluation challenges, which are many. These include evaluating accuracy and coherency. While it is relatively easy to assess accuracy automatically, in the majority of instances coherency requires manual interventions, and this will only be exacerbated. Furthermore, concerns about coverage and succinctness are also relevant to input as well as output, and should be included as a consideration. This is a similar problem to that experienced in crowd-sourced KGs such as Wikidata [1], where variations in the input (such as background and skill of contributors) also demands evaluation as much as the output.

Regarding the evaluation of LLM outputs, one interesting suggestion was made related to using adversarial algorithms. Recently, an adversarial attack on BARD (now Gemini) [20] asking to repeat

one word iteratively eventually caused the LLM to reveal gigabytes of its training data, demonstrating the usefulness of such processes in testing systems [45]. Another proposed aspect of evaluation was human satisfaction. Other research has shown that the natural language element of LLMs, often read by humans as possessed of emotion, can make people feel that the process of using the LLM was successful even if the end result was not [63]. This could be potentially problematic for KE, which requires logic over feelings. A comparable, well-researched situation is that of emotion in financial markets, which can move prices up or down based on public mood rather than underlying economic fundamentals.

Prompting emerged as a key topic across all areas. Although many interviewees spoke of how they found previous skills useful in developing effective prompts (and equally, that it was very possible to use ineffective prompts by accident), some interviewees also felt that prompting skills were not important. However, ultimately prompting was utilised for all the different KE tasks. This flags a new era and requirement for a new skill set in the field of KE, with parallels to the introduction of crowdsourcing to KG construction around a decade ago. Using crowdsourcing means for large-scale KG construction was a successful innovation supporting many intelligent applications today.

Today, building a KG requires a domain expert, knowledge engineer and ontology engineer (although 2 or more people may play these roles). When considering the design of LLM-driven support for KGs, it is key to establish new workflow patterns and templates so that KEs are able to effectively exploit aspects of LLM without pitfalls, such as poor prompting leading to ineffective results. This suggests the role of prompt engineer may be a useful addition.

One of the great benefits of generative AI is its ability to work multimodally, although these are expensive and hard to use in real life scenarios. Even for non-specialised generative AI such as ChatGPT output can be in the form of text, images, graphs, or programming code. There have been great advances recently in generative video AI, such as SORA. However, this variety of formats of output is difficult to process for KE without human intervention. Furthermore, the humans involved have to possess the skills in for working in multiple modalities. While some of our respondents found images to stretch their skills, this could be many times more demanding when working with audio or haptics. Further, we may see the emergence of new methodologies in which domain experts, engineers and LLMs co-operate for more effective elicitation of requirements, quality assessment and scalability of KE processes. There are existing examples of LLMs supporting users via copilot approaches, such as those used by Github to support developers by generating code or by Microsoft to support users in search and document exploration. This approach appears to have merit.

5.0.1 Key Insights.

- Effective prompting skills are a key requirement. These are more easily acquired by those who have worked in natural language processing. Others, particularly knowledge engineers, have less training in this area so more support and development is required.
- Further research is required into (i) sharing prompting guidelines or templates that have been used successfully, and (ii) effective multimodal prompting.

- One size may not fit all for using LLMs for KGs. Horizontal generative AI such as ChatGPT are appropriate where wide-ranging content is required, but a vertical, specialised generative AI may be more effective in, for example, preventing syntactical errors.
- Copilot approaches may support multimodal and quality assessment work.

6 CONCLUSION

6.1 Conclusion

In this study we describe the use of a hackathon – an open innovation tool – to explore the early promise and challenge in the use of LLMs to address key issues facing KE. We present the views and experiences of hackathon participants using LLMs to augment KGs. Although in theory LLMs offer potential to address some of the key challenges in KE, this is not an unmixed solution. LLMs can extract not only concepts but also relationships, constraints, parameters and more, which is very beneficial for knowledge acquisition and requirements elicitation. They also proffer facility in working with multiple types of content. Recent advances in “large action models” that can interact with web interfaces to perform online activity promise even further integration opportunities for multimodal output [54]. In particular, if LLMs can assist humans to perform evaluation faster (or in an entirely novel manner) and with more objectivity, this offers a real opportunity for improvement on the current state of play. However, there is an “abundance of caution” amongst the hackathon participants we interviewed regarding the use of LLMs in KE. This was derived not only from concern over the well-documented accuracy and trust issues inherent in LLMs, but also because they felt that some issues – such as those of evaluation – would be hard to address in an automatic fashion if the problem had not first been solved manually. Lastly, there was substantial awareness that skill in prompting was required in order to effectively interact with LLMs.

6.2 Limitations and Further Work

The participants in the hackathon were primarily PhD students based in labs in Europe, and their approaches to these issues are informed by this background. The challenges in the hackathon were set (as is customary) by the hackathon organiser, and therefore represent their views on the important aspects of LLM/KG use to investigate. Further, the analysis above is based on a subset of the total data we collected, which will eventually enrich the insights further.

The above are early results of a more extensive inquiry that seeks to also explore what skills are required in order to successfully use LLMs for KE tasks and how aware are KEs of bias as a safety issue in LLM use for KE.

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A INTERVIEW TOPIC GUIDE

Consent. Consent text (to read to interviewer):

With this research, we aim to investigate how knowledge engineers and practitioners use LLMs to develop and maintain knowledge graphs, ontologies, and knowledge bases. We will not collect any personal data. The responses to the questions are fully anonymous. The collected data will be shared with otter AI, an automated transcription service hosted on an AWS server in the US. Your data will be processed under the terms of UK data protection law (including the UK General Data Protection Regulation (UK GDPR) and the Data Protection Act 2018). Transcripts will be kept until the publication of the research.

If you consent to participate in this study, please say loud and clear “Yes I consent”.

Introduction. Give a bit of an intro of what are the main topics of the questions (e.g., your experience during Hackathon, challenges, skills, automation, evaluation, responsible AI). In the end we will ask you about anything important you want to mention, and you will have time to add comments. Mention that there is no wrong answer.

Questions.

- (1) What is your experience with knowledge engineering and building knowledge graphs/ontologies?
- (2) How did you find the process of interacting with LLMs to solve knowledge engineering tasks?
- (3) We found that during the hackathon it was challenging for the groups to (i) find a dataset for testing their pipelines, (ii) find the right prompt questions for the LLMs, and (iii) evaluate the quality of each step toward building an ontology. Which one do you think is the most important?
- (4) What do you think were the most important skills you have to complete the tasks during the Hackathon?
 - (a) Do you feel you had those skills, or do you need to gain experience to complete several tasks?
- (5) Was there any automated process in the project you worked on during the Hackathon?
 - (a) What was the purpose of this automation?
- (6) Currently, to evaluate ontologies and knowledge graphs constructions, we use metrics like precision and recall against gold standards, or human evaluation (semantic metric). Do you think this is enough?
 - (a) What they would look like new benchmarks and metrics?
- (7) Have you previously used safety testing for responsible AI? Safety testing can be Bias testing against discrimination,

Ethical considerations, risk assessment, test for long-term effects.

- (a) Do you think this can be used in the LLMs knowledge engineering scenarios or we need to develop specific solutions?

- (8) Do you think there is anything else important for us to know related to LLMs and knowledge engineering?